Article

MA-SQLGrid: A Robust Multi-Agent Framework for Text-to-SQL in Power Grid Databases

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Featured Application

MA-SQLGrid is an experimental read-only decision-support framework for auditable natural-language access to structured power-grid records. Generated SQL and returned source rows require human inspection before consequential use; no operational control, protection, or maintenance decision has been validated.

Abstract

Natural-language access to power-grid databases requires traceable schema grounding, database-enforced read-only execution, and explicit handling of missing evidence. We operationalize "robust" only as tested mutation denial, bounded execution, complete-evidence gating, and constructed-witness behavior; universal or deployment robustness is not claimed. We present MA-SQLGrid, a five-role software architecture coordinated through an append-only blackboard and deterministic adjudicator. Its implemented boundary combines SQLite read-only mode, `query_only`, an authorizer, disabled extensions, resource limits, and fail-closed named-state coverage. Earlier GridDB and BIRD experiments evaluate frozen prompting workflows rather than autonomous multi-agent generation. A separate 700-call study uses a prospectively frozen call-and-selection procedure on development-visible GridDB items to test presented-value and reference-free selection components. A no-generation release-v3 re-execution applies three selectors to a shared 5760-attempt collection for eight historical candidates on 180 GridDB questions. Because frozen pre-run tests accessed same-item gold-derived outcomes, this release is descriptive. Fixed order, validation-only ranking, and complete three-witness selection obtained 80/180, 100/180, and 101/180. The two adjudicating selectors tied at the highest score on 130/180 questions; reversing candidate order produced 117–118/180. Their sole difference, Q039, is a constructed and semantically ambiguous projection trace. The evidence supports a reproducible architecture with tested read-only execution and bounded-evidence mechanisms, but not five-role superiority, qualified power-grid semantic validity, deployment safety, or universal robustness.

Keywords: text-to-SQL; multi-agent systems; power-grid databases; schema grounding; execution validation; constructed-state testing; reproducibility

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1. Introduction

Power-grid organizations maintain relational data about assets, locations, topology, sensor readings, work orders, technicians, and maintenance activity. SQL provides precise access to these records, but it requires knowledge of table names, join keys, identifiers, units, aggregation conventions, and local coding rules. Text-to-SQL systems aim to reduce this access barrier by translating a natural-language request into an executable query. In an engineering workflow, however, a fluent query is not sufficient evidence of correctness. A query can execute while selecting the wrong status field, omitting a required filter, using an unintended join path, or returning the expected number of columns with the wrong content.

The risk is amplified by the semantics of grid records. An identifier that looks numeric may be a categorical code; a timestamp can refer to sensing, issuance, acknowledgement, or closure; and an asset status may be current, event-time, or work-order-specific. The same user wording can therefore map to several executable queries. Natural-language access must expose schema and value grounding, preserve the generated statement, and retain the returned source rows so that an engineer can inspect the interpretation. It should also abstain when the available evidence cannot distinguish candidates.

Cross-database benchmarks such as Spider and BIRD have made schema generalization and realistic database content central evaluation concerns [1,2]. Schema-aware models, constrained decoding, context selection, decomposition, and execution guidance address different failure surfaces [3–7]. These methods also expose a design problem: generation steps are frequently bundled together. A system may change schema serialization, value presentation, domain normalization, structural hints, candidate count, and repair policy simultaneously. An observed score difference then cannot be assigned cleanly to schema linking, reasoning, or validation.

Recent Applied Sciences studies illustrate the breadth of this design space. RGISQL refines grammatical information in a relational-graph model, while zero-shot prompting work compares prompt strategies and benchmark behavior [8,9]. Schema-retrieval systems use embeddings and vector stores to limit context, and domain-specific work such as DKA-SQL adapts knowledge to power-grid questions [10,11]. These contributions focus primarily on generation quality or contextualization. The present work instead asks what can be established when candidate production, validation evidence, state-based diagnostics, and deterministic selection are recorded as separate stages.

The original MA-SQLGrid concept addresses this problem through five specialized roles. A Query Analyst interprets the request, a Schema Cartographer maps it to the database, an SQL Synthesizer packages candidate queries, a Validation Engine checks safety and execution, and a role historically named the Counterfactual Critic—operationally a Metamorphic-State Critic—records evidence from named constructed database states. A separate deterministic adjudicator resolves the resulting evidence. This separation makes the trace inspectable: a reviewer can distinguish what the analyst inferred, which schema elements were retained, which candidates were proposed, which candidates executed, and why a final candidate was selected or rejected.

The architecture must nevertheless be separated from the evidence used to evaluate it. The inherited GridDB factorial study was not a multi-agent execution. It used one model generation per question and prompt cell, followed by deterministic parsing, read-only validation, execution, and offline scoring. The BIRD study likewise compared four frozen prompting procedures rather than five interacting agents. These assets can motivate modules, supply baselines, and test replay interfaces, but they cannot be renamed as results of the new coordination core.

This evidence distinction is especially important for candidate-pool selection. If eight historical candidates already contain a correct query, a selector may recover it without improving generation. If the candidates originate from different models and prompt packages, the pool also receives more physical calls than a one-shot baseline. Conversely, a sophisticated selector cannot repair a pool that contains no adequate query. Generation coverage, safe executability, reference-free ranking, and gold-relative accuracy are therefore reported as different endpoints.

The research gap is consequently not another unqualified agent count. It is an evidence architecture that permits a future matched experiment to replace one role at a time while retaining the same message schema, executor, state requirements, physical-call accounting, and gold boundary. The present reconstruction supplies that architecture and reports what existing assets can and cannot say before such an experiment is run.

This distinction leads to three questions. First, does the implemented five-role software conform to its typed handoff, database-enforced, gold-isolated, and fail-closed contracts? This is a software-conformance question, not an efficacy comparison. Second, what do the completed experiments establish about context packages, presented values, public-database transfer, and constructed-state agreement? Third, when generation is held fixed at eight already frozen candidates per question, how do fixed-order, validation-only, and complete-three-witness selectors behave under the same evidence-collection operations? The third question estimates historical-pool offline selection only; schema grounding is retained as trace evidence and no five-role end-to-end, new-generation, dialogue, or deployment effect is estimated.

The analysis answers these questions through a hierarchy rather than one headline accuracy. Software tests address mutation denial, resource termination, trace completeness, and fail-closed coverage. Frozen experiments address prompt-package and model behavior. The public BIRD protocol tests portability outside the grid domain. The release-v3 re-execution addresses deterministic selection from a fixed historical pool, with prior same-item outcome access disclosed. This hierarchy prevents a passed executor test from being presented as semantic correctness and prevents a descriptive selector result from being presented as prospective multi-agent superiority.

The paper makes four contributions. First, it defines and tests software conformance for a five-role coordination core, a database-enforced SQLite executor, and a deterministic adjudicator whose required counterfactual coverage fails closed; no experiment estimates a five-role benefit. Second, it conservatively integrates a 1440-prediction GridDB factorial experiment, an audited 700-call component study with a prospectively frozen procedure on development-visible items, a 25,920-row multi-state ledger, and a 5000-call BIRD comparison while retaining their original identities and evidence classes. Third, it reports a deterministic descriptive re-execution over the existing eight-slot pool using three query-blind constructed witnesses with distinct database hashes; all methods choose only after the same shared 32-attempt evidence collection per question is complete. Fourth, it publishes complete counts and adverse diagnostics: 130/180 top-score ties for both adjudicating selectors, approximately 5.4 candidates per top-tie set, all 18 order/weight/threshold cells, and the semantically ambiguous 1/180 Q039 trace.

No Spider or WikiSQL experiment is reported, no unsupported large power-grid question–SQL corpus is asserted, and the offline result is not relabeled as autonomous multi-agent generation. In the retained title, *multi-agent* names the typed five-role software decomposition, not five concurrently hosted language models, and *robust* names only the explicitly tested mutation-denial, bounded-execution, complete-evidence, and three-witness mechanisms. The title is therefore a framework identity rather than an empirical superiority claim. Qualified domain review and a matched new-generation experiment remain mandatory gates before broader interpretation.

The remainder of the paper first positions the framework relative to schema-linking, agentic, and grid-data research. Materials and Methods then define the trust boundary, artifact chronology, five role contracts, adjudication rule, and six evidence sources. Results retain every adverse diagnostic, including order sensitivity, top-score ties, and a semantically ambiguous trace. The Discussion interprets only claims licensed by those designs and separates reproducible software behavior from operational readiness.

2. Related Work

2.1. Text-to-SQL Evaluation and Schema Generalization

Text-to-SQL systems map a natural-language question and a database schema to a query. Spider established a cross-domain setting with previously unseen databases, drawing attention to table selection, joins, grouping, nesting, and ordering [1]. BIRD extends public evaluation toward larger and more content-rich databases [2]. A benchmark result depends on the exact split, evaluator, database engine, execution boundary, and model configuration. We therefore report only the completed BIRD Mini-Dev protocol and make no claim about public datasets that were discussed but not run.

Exact SQL match and execution evaluation answer different questions. String match can penalize equivalent SQL, whereas result equality can reward nonequivalent queries on an insufficiently discriminating database state. A robustness evaluation should retain predicted SQL, record every failure, and specify the state on which equality was observed. MA-SQLGrid uses strict result equality for inherited snapshot analyses and separately reports constructed-state agreement and bounded metamorphic invariance. Neither auxiliary endpoint is semantic certification.

Execution equivalence is particularly vulnerable when a table is small, a predicate happens not to filter any row, or two joins coincide on one snapshot. Test-suite accuracy and multiple database states can reduce this accidental agreement, but their value depends on how states are constructed and whether their transformations preserve the intended query semantics. Our prediction-blinded state suite is therefore reported as a constructed diagnostic. The three release-v3 witnesses are even narrower: they test invariance to irrelevant relations, storage/index changes, and nullable extensions, not correctness under realistic grid evolution.

Work on SQL-oriented language-model teams shows another distinction between using several models as options and coordinating explicit roles [12]. A team can increase candidate diversity, whereas an agent architecture can formalize ownership of intent analysis, schema grounding, generation, validation, and critique. Either may help, but a fair benefit claim requires equal model calls, candidate counts, and selection information. The current paper supplies the role contracts and a fixed-pool diagnostic, not that matched end-to-end comparison.

2.2. Schema Linking, Context Packages, and Output Contracts

Schema linking connects question spans to tables, columns, values, and foreign-key relations [3,13,14]. Full-schema context preserves recall but can increase token cost and distraction. Compact context reduces the search space but risks dropping a required join key or predicate. Value examples can disambiguate coded attributes, while question-derived instructions can identify aggregation, result shape, ordering, or a limit.

Retrieval-based schema selection frames this trade-off as a recall problem over relevant tables, columns, and relationships [10]. In a grid database, lexical relevance alone may miss bridge tables or coded fields whose names do not occur in the question. A safe compact package should retain foreign-key closure, report unmatched question spans, and abstain when required relations cannot be justified. The current Cartographer records these elements, but its lexical rules are an inspectable baseline rather than a learned retrieval contribution.

The inherited GridDB factorial study isolates two implemented prompt factors: a full-DDL/global-values package versus a compact/domain-grounded package, and absence versus presence of a composite structural/SQL-operation hint. The compact package is not a pure schema-length manipulation because it also changes value presentation and adds corpus-tailored normalization. The hint is not merely formatting because it can include aggregation, grouping, projection, ordering, and limit guidance. We retain these precise labels instead of assigning broader causal interpretations.

Constrained generation and execution guidance reduce invalid candidates. PICARD constrains decoding so inadmissible SQL continuations can be rejected during generation [4]. CHESS combines contextual grounding and SQL synthesis components [7]; other prompt-based methods investigate decomposition and representation choices [5,6,15]. MA-SQLGrid differs at the coordination boundary: its new core accepts externally produced candidates, checks them with deterministic contracts, and records evidence before selection. It does not claim that the current lexical cartographer supersedes learned schema-linking systems.

This boundary also changes the reproducibility unit. A monolithic prompt often records only the final SQL, while the blackboard records the question-only intent, schema links, canonical candidates, safety decisions, execution hashes, named-state coverage, and final rule outcome. Intermediate records do not guarantee correct reasoning, but they allow an audit to locate whether a failure arose before generation, at parsing, under database authorization, during evidence collection, or at adjudication.

2.3. Agentic Decomposition and Auditable Control

Agentic decomposition can make intermediate decisions explicit, but naming pipeline stages as agents does not establish a multi-agent benefit. Such evidence requires a comparison that holds the model, data, decoding configuration, candidate count, and physical generation budget constant. Otherwise, a multi-candidate system can improve simply because it receives more samples.

Multi-agent systems are also used in energy management, where agents represent physical or decision-making entities and exchange state or control information [16]. That tradition differs from language-model role decomposition. MA-SQLGrid uses agents as typed software responsibilities over an information trace; it neither controls grid equipment nor models autonomous market or device actors. Making this distinction explicit avoids importing validation claims from one meaning of "multi-agent" into another.

MA-SQLGrid treats an agent as a role with a typed contract rather than assuming that every role must be a separately hosted model. The Analyst and Cartographer can be deterministic or model-assisted; the Synthesizer can receive candidates from a frozen generator; the Validator uses a fixed read-only policy; the Critic consumes named-state evidence; and the Adjudicator is deterministic. This design allows components to be replaced without changing the blackboard schema and prevents gold-guided selection from being hidden behind an opaque debate label.

Debate and self-repair designs can create useful diversity, but they also introduce stopping rules, message-history effects, and additional sampling opportunities. Reproducible evaluation should record which role caused each physical model call, whether a retry replaced or supplemented a failure, and which information was visible at each turn. MA-SQLGrid uses an append-only trace and zero hidden retries for this purpose. Its current offline study avoids new calls entirely, so it tests controller interfaces without estimating conversational benefit.

2.4. Power-Grid Data and External Validity

Power-system datasets combine topology, equipment attributes, measurements, time series, costs, and constraints. RTS-GMLC and SimBench provide reproducible engineering resources that can be transformed into relational structures [17,18]. Their availability does not automatically create a text-to-SQL benchmark: intended projections, units, ordering, ties, and reference SQL require semantic review. The current RTS-GMLC, SimBench, and NERC-derived candidates are consequently machine-adjudicated silver resources, not expert gold.

DKA-SQL is a direct power-grid text-to-SQL precedent [11]. Its published corpus, method, models, and evaluation boundary differ from ours. Because no official implementation was used, the current BIRD protocol is neither a DKA-SQL reproduction nor a head-to-head comparison.

External validity has two axes. Database structure must resemble target assets, topology, measurements, and maintenance records, while questions and reference answers must reflect how qualified users formulate and resolve information needs. RTS-GMLC and SimBench contribute the first axis but not automatically the second. BIRD contributes a public language-to-database benchmark but is not grid-specific. GridDB supplies a controlled conjunction of both axes, yet remains synthetic, development-visible, and without completed independent expert adjudication.

Table 1 summarizes the closest methodological axes. The comparison is architectural rather than a score table because datasets, call budgets, and evaluators differ. It shows that the present novelty lies in combining typed role traces with database-enforced execution and complete named-state eligibility, while the strongest missing evidence is a prospective, title-concordant, expert-reviewed evaluation.

Table 1. Positioning against adjacent Text-to-SQL approaches; entries describe reported design emphasis, not comparable performance.

Approach	Principal emphasis	Evaluation setting	Difference from MA-SQLGrid
RGISQL [8]	Grammatical and relational graph modeling	Public Text-to-SQL tasks	Learned generator rather than a traced executor/adjudicator boundary
Zero-shot prompting [9]	Prompt strategies and LLM behavior	Public benchmarks	Prompt comparison without the five-role blackboard
Schema retrieval [10]	Embedding/vector-store context selection	LLM SQL generation	Retrieval quality rather than fail-closed named-state selection
DKA-SQL [11]	Domain knowledge adaptation	Power-grid Text-to-SQL	Different corpus and implementation; not reproduced here
MA-SQLGrid	Typed trace, read-only executor, deterministic selection	GridDB case study plus BIRD portability	Prospective five-role benefit and expert semantic validation remain open

3. Materials and Methods

3.1. Task and Evidence Boundary

For question x_i, read-only database D_i, and introspected schema S_i, a generator produces one or more candidates

$$\mathcal{Y}_i = \{\widetilde{y}_{i1}, \ldots, \widetilde{y}_{ik}\}. \quad (1)$$

Each candidate must satisfy a single-statement read-only contract. The accepted leading form is `SELECT` or `WITH`; comments, multiple statements, write operations, schema changes, attachments, and pragmas are rejected. A candidate is eligible for adjudication only when it is safe and executable.

The offline evaluator may later compare the selected query with a reference result. Let $E_i = 1$ when the selected SQL executes and its result equals the frozen reference result under the registered evaluator, and $E_i = 0$ otherwise. Reference SQL, reference results, E_i, and any derivative correctness field are forbidden from all five agents and the Adjudicator. Evaluation occurs only after the blackboard is sealed.

This boundary yields three distinct objects. A candidate is *admissible* when its syntax and requested operations satisfy policy, *executable* when it completes within the registered database limits, and *correct under the frozen evaluator* when its result matches the reference after sealing. Admissibility is neither executability nor correctness. The manuscript preserves all three labels because collapsing them would let an executable but semantically wrong query appear validated.

The framework's response contract contains the selected SQL, stable candidate and source identifiers, referenced tables and columns, execution-state and result hashes, state-coverage summary, abstention or selection reason, and the sealed trace digest. It does not present generated text as an authorization decision. A calling application must separately authenticate the user, restrict the database view, and decide whether displaying the returned rows is lawful. Those controls lie outside the evaluated framework.

3.2. Data Resources and Provenance

GridDB-Maintenance-v2 v0.1 is a synthetic SQLite case study with eight tables and 98 rows: `asset_types` (6), `assets` (18), `grid_topology` (9), `locations` (8), `maintenance_logs` (8), `sensor_readings` (26), `technicians` (8), and `work_orders` (15). It contains 200 natural-language–SQL records: 20 development records and 180 factorial evaluation records. The evaluation records include 66 easy, 91 medium, and 23 hard items and map to 70 normalized gold-SQL structural clusters, of which 58 are singletons. The evaluation partition had been visible during project development; it is a controlled finite-set case study, not a sealed production benchmark.

BIRD Mini-Dev contributes 500 public items over 11 SQLite databases. The authorized protocol froze item and method order, prompts, two local model files, evaluator behavior, Python 3.10.11/SQLite 3.40.1, and zero retries. Each backbone made 2500 calls and produced 2000 final predictions. An independent audit re-executed all 4000 predictions and 500 gold queries with zero score mismatches. Two earlier failed attempts, totaling 2476 physical calls, remain immutable incidents and are excluded from scores.

The GridDB reliability release contains 18 states: the canonical snapshot, three insertion permutations, and 14 additional schema-valid stress states. The scorer produced 25,920 prediction-state rows (1440×18). A prediction-blinded automated protocol admitted a 66-question subset for the primary 15-state analysis and retained 114 order-sensitive questions as diagnostics. These constructed states are not operator-certified grid snapshots.

Artifacts are labeled Inherited, Recomputed, New, or Diagnostic. The coordination core is implementation evidence and its replay is Diagnostic. Neither is a New execution-accuracy result.

The evidence classes also preserve chronology. "Inherited" means that predictions and primary outcomes predated the current coordination framework. "Recomputed" means that a retained input was processed again without a new generation claim. "New" is reserved for an operation whose inputs, rule, and visibility boundary were frozen for that question before its outcome was accessed. "Diagnostic" permits post hoc inspection but not confirmatory language. These labels are assigned per result, not per manuscript, because one paper can legitimately contain several evidence classes.

Table 2 separates resources by domain, evidence class, visibility, and semantic-review status. BIRD is explicitly non-grid. RTS-GMLC and SimBench provide authentic power-system structures, but their question–SQL pairs remain machine-generated silver candidates with zero completed qualified-human reviews and zero sealed items.

Table 2. Data-resource scope and evidence status. Counts describe retained local artifacts; they are not interchangeable benchmark denominators.

Resource	DBs	Tables	Rows	NL–SQL	Evidence class	Visibility and review status
GridDB v0.1	1	8	98	200	Synthetic grid; inherited development case	20 dev + 180 evaluation; development-visible; no documented independent dual-expert review
GridDB state suite	18 states	8	state-dependent	0 new	Constructed grid-state diagnostic	Prediction-blinded state construction; not operator-certified
RTS-GMLC pilot	1	10	360,530	55	Authentic grid structure; machine silver	`AUTO_CANDIDATE`; 0 human-reviewed; 0 sealed; source-use notice applies
SimBench pilot	1	8	874	36	Authentic grid structure; machine silver	Development-visible; 0 human-reviewed; 0 sealed; ODbL/DbCL review required before redistribution
BIRD Mini-Dev	11	dataset-specific	dataset-specific	500	Public non-grid portability benchmark	Frozen public split; BIRD v1.1 protocol; not evidence of power-grid validity

3.3. Five-Agent Coordination Framework

Table 3 gives the master protocol-to-estimand map used throughout Results. It prevents denominators, visibility histories, and inferential licenses from being pooled across studies.

Table 3. Master protocol, estimand, visibility, and claim-license map. "Composition interval" denotes resampling sensitivity of the observed finite corpus, not a population confidence interval.

Protocol	Unit and *N*	Dependence proxy	Generation budget	Endpoint	Gold visibility	Status/multiplicity	Strongest licensed claim
GridDB factorial	question; 180/cell	70 normalized-SQL groups	1 call/item/cell	strict execution equality	development-visible items; scored after generation	finite-set; 9-test Holm family	prompt-package behavior, not five-role benefit
Component E1/E2	question; 170 paired or 180	61 or 70 groups	700 total calls	execution equality	procedure frozen before calls; items development-visible	composition intervals; registered families	component behavior only
Formal-v5 states	question; 66 primary	12 structural groups	0 new calls	15-state logical AND	state construction prediction-blind; gold-relative scoring later	exact 2^{12} signs; 9-test Holm	constructed-state agreement
BIRD v1.1	item; 500 across 11 databases	database	B0–B2: 1; B3: 2 calls/item	execution accuracy	public benchmark; evaluated after calls	12 pairwise tests with Holm	non-grid workflow portability under unequal calls
Replay	question; 180	none	0 new calls	pool coverage	prior evidence existed	descriptive	interface feasibility only
Release v3	question; 180	none	shared 5760 evidence attempts	historical-pool execution match	same-item derived outcomes previously accessed	outcome-exposed descriptive sensitivity	deterministic selector behavior, not coordination efficacy

The evidence chronology is shown in Table 4. Release v3 is classified as a post-audit descriptive re-execution: its execution path sealed decisions before directly loading raw gold, but the frozen regression suite had already read v2 gold-derived outcomes for the same 180 items. Consequently, hashing and within-run sealing provide traceability, not prospective outcome blindness.

Table 4. Study chronology, visibility, and permitted interpretation.

Study	Data/candidate visibility	Rule or analysis freeze	Outcome access	Permitted interpretation
GridDB factorial	Evaluation partition visible during development	Original protocol before generation; v3 hierarchy is post-review reanalysis	After generation	Inherited finite-corpus prompt-package evidence
Component study	Candidate prompts frozen before calls	Component protocol frozen before calls	After blackboard/selection seal	Inherited component evidence; not five-role coordination
Formal-v5 states	States built prediction-blind	Scorer frozen before formal scoring	Gold-relative scoring after state freeze	Constructed-state diagnostic
BIRD v1.1	Public Mini-Dev	Protocol/runtime frozen before 5000 calls	Formal evaluation after calls	Non-grid workflow portability; unequal calls by method
Retrospective replay	All predictions and T0 evidence already existed	Diagnostic logic hash-locked	No gold accuracy computed	Candidate-pool coverage only
Descriptive re-execution v3	Eight historical candidates and their prior evaluations already existed	Question-only selection view, three distinct witnesses, construction code, tests, and rules hash-bound in v3; frozen tests had read same-item v2 outcomes	Raw gold directly loaded by the v3 execution path after all 180 boards were sealed	Reproducible descriptive selector behavior on previously evaluated items; no prospective, five-role, or generation-effect claim

MA-SQLGrid comprises five specialist roles coordinated by a deterministic controller. Figure 1 is derived from the implemented source rather than an aspirational free-form agent diagram. The controller is not a sixth reasoning model: it orders calls, applies the registered rule, and seals the trace. Messages receive consecutive sequence numbers, and the complete trace is serialized canonically to a SHA-256 digest.

The **Query Analyst** produces a structured question-only intent record containing lexical tokens, detected aggregations, ordering requirements, and an explicit limit when present. It does not select SQL or see correctness. The **Schema Cartographer** maps lexical intent to tables, columns, and foreign-key edges. The current skeleton uses deterministic overlap and records unmatched tokens; a richer GridDB adapter adds disclosed domain rules and graph expansion.

The **SQL Synthesizer** packages externally produced SQL strings. It contains no model client, performs no hidden API call, canonicalizes whitespace and semicolons, removes exact duplicates without changing first occurrence, and assigns stable identifiers. The **Validation Engine** separates lexical admissibility from the executor trust boundary. The executor bound into release v3 opens a frozen database through a URI with `mode=ro&immutable=1`, enables SQLite `query_only`, disables loadable extensions, and installs an authorizer that admits selection, reads, selected functions, and recursive-query opcodes. Writes, DDL, transactions, attachments, pragmas, metadata enumeration, dangerous file/extension functions, and reads outside configured table/column allowlists are denied. A progress handler enforces registered time and opcode limits, and result fetching enforces a row cap. Every attempt records database and SQL hashes, authorization or resource failure, elapsed time, row count, and result hash; no automatic retry removes a failure.

After R3 review, a separate FINAL executor variant adds explicit function allowlisting, maximum projected width, a raw scalar-cell byte boundary, and a budget for the total returned result. BLOBs and memory views are charged at their original payload length before their public representation is replaced by a digest; the total budget charges the original payload plus deterministic row, column, type-envelope, delimiter, and container

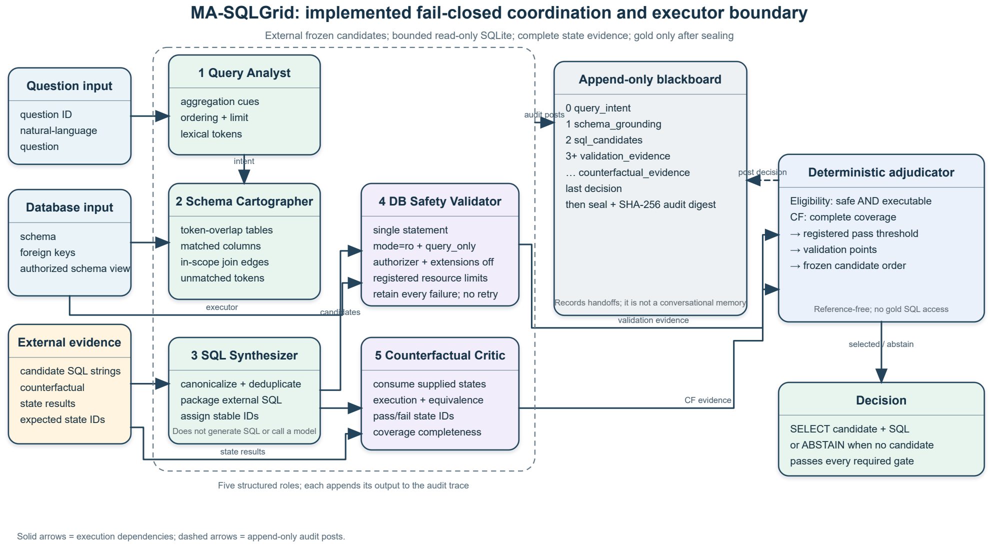

Figure 1. Implemented MA-SQLGrid boundary (code-native SVG and deterministic raster derivative). Candidate SQL is supplied externally. The database validator enforces read-only and registered resource limits; counterfactual-required selection demands complete named-state coverage; gold SQL remains outside the blackboard until sealing. User authorization and operational approval remain outside this software boundary.

overhead. Fourteen additive tests pass, including 1, 5, and 50 MB `zeroblob` denial under a 1 KB cell limit, an independent total-result-budget denial, exact boundary behavior, trace retention, oversized text/result denial, and wide-projection denial. These FINAL controls were implemented after release v3 and were *not* used to obtain its 5760-attempt ledger or 80/100/101 counts. A SQL table/column allowlist is a query-scope mechanism supplied by the caller; it is not authentication, identity management, row-level access control, or proof that a particular utility user is entitled to the data. Both historical and FINAL variants therefore establish tested SQLite mutation denial and explicitly scoped execution bounds only, not user authorization, data minimization, process isolation, or operational approval.

The executor threat model assumes an untrusted candidate SQL string but a trusted host application, trusted SQLite library, correctly configured database allowlist, and immutable local database file. It aims to prevent mutation through SQL, restrict visible relations, and terminate selected resource-exhaustion patterns. It does not defend against a compromised Python process, operating-system escape, side channels, malicious SQLite extensions already built into the runtime, or authorization mistakes made before the call. Production use would require a separate process or container, least-privilege database views, authenticated user context, logging policy, and independent security review.

Failure handling is intentionally fail-closed and non-retrying. A parse denial, authorizer denial, timeout, opcode limit, row limit, width limit, cell limit, or total-result limit creates a retained attempt record and makes that candidate ineligible for the current decision. The controller cannot silently weaken a limit or substitute a repaired query. A repair procedure may be studied as a separate registered generation step, as in BIRD B3, but its extra physical call and visibility must remain explicit.

The **Metamorphic-State Critic** (the historically named Counterfactual Critic) consumes named state results, rejects duplicate state identifiers, and records evaluated, passed, failed, and missing states. When state evidence is optional, missing evidence remains diagnostic and does not break prior no-state behavior. When the caller requires state evidence, coverage must equal the complete registered state set and the pass threshold must be met; otherwise the candidate is ineligible. Thus an incomplete 1/1 record cannot outrank a complete 10/11 record. Existing formal-v5 labels compare predictions with gold and remain forbidden during selection.

The blackboard is append-only at the application layer: a role posts a typed message and cannot rewrite an earlier message. The controller checks role, message type, sequence number, and required fields before acceptance. Canonical serialization fixes key order and whitespace before hashing. This makes two runs comparable at the trace level and exposes an unexpected hidden field or changed handoff. It does not prevent a malicious host process from altering memory or files; process isolation and remote attestation are outside scope.

Information flow is intentionally asymmetric. The Analyst receives the question but no candidate SQL. The Cartographer receives the question, intent record, and introspected schema. The Synthesizer receives external candidate strings and assigns identities but cannot execute them. The Validator receives one candidate plus a scoped executor. The Critic receives named-state outcomes but no reference correctness. Only the deterministic Adjudicator receives the eligible evidence summaries. Gold enters a separate offline evaluator after the decision message and board digest exist.

3.4. Deterministic Adjudication and Abstention

Eligibility always requires safety and successful execution. The archived 100-point representation assigns 40 points for each of those two properties, followed by 10 for question-derived aggregate-shape conformity, 5 for question-derived ordering conformity, and up to 5 for lexical value hits. Because every ranked candidate has already passed both hard gates, the first 80 points are constant and contain no ranking information. The effective evidence score is therefore $e(y) = 10I_{\text{shape}}(y) + 5I_{\text{order}}(y) + 5V(y)$ for an eligible candidate y, with $V(y) \in [0,1]$. In no-counterfactual mode, the controller selects an $\arg\max e(y)$ and ties preserve frozen candidate order. In required-state mode, incomplete named-state coverage or failure of the registered pass threshold makes y ineligible before the same ranking. If the eligible set is empty, the controller abstains. This piecewise gate–score–tie function is mathematically equivalent to the archived implementation; it exposes that safety/execution are eligibility controls rather than discriminative score evidence. V3 does not establish that the weight design preceded access to derived outcomes for the 180 items, so weight, pass-threshold, and reversed-order results remain descriptive sensitivity analyses.

The archived 40/40/10/5/5 allocation—equivalently, hard safety/execution gates followed by a 10/5/5 evidence score—is an illustrative deterministic policy, not a learned, calibrated, or theoretically optimal weighting and not an algorithmic performance contribution. Its formal invariants are: an unsafe or non-executable candidate is never selected; required named-state coverage is complete before its threshold is evaluated; gold-derived correctness is absent from the decision record; equal scores resolve by the frozen order; and an empty eligible set yields abstention. These invariants support conformance testing and replay determinism. They do not imply semantic correctness, calibrated confidence, or superiority over a simpler policy.

Question-derived shape checks are deliberately weak. They detect cues such as "how many", "highest", "first", and explicit limits and compare them with the projected aggregate or ordering structure. They do not infer business semantics such as whether "active" refers to an asset, alarm, or work-order state. Lexical value hits likewise reward only presented values traceable to the question and context; they are not evidence that the associated column is the intended one. These low-bandwidth features keep adjudication inspectable but explain why large tie sets remain possible.

Abstention has two layers. Pre-ranking abstention occurs when no candidate passes safety and execution requirements or when required state coverage is incomplete. Post-ranking ambiguity is currently resolved by stable order rather than abstention because no pre-registered margin threshold exists in v3. The high tie rates observed later show that future work should freeze an ambiguity policy—for example, abstaining on unresolved top ties or requesting a user clarification—before examining new outcomes.

For this paper, robustness is a vector rather than a universal label (Table 5). Only tested components are reported as evidence.

Table 5. Tested and untested robustness dimensions and their evidence boundaries.

Dimension	Registered operation/end point	Evidence in this paper	Not established
Mutation safety	DB-enforced denial of write/DDL/attach/pragma operations	Adversarial executor tests; unchanged database hash	User authorization or whole-process sandboxing
Resource boundedness	Timeout/opcode, row, raw-cell-byte, budgeted-result-byte, and output-width termination	Recursive-query, row/width overflow, oversized text, and 1/5/50 MB BLOB tests with retained failure traces; FINAL limits are not retroactive to v3	Worst-case process-memory bounds for every SQLite feature
Evidence completeness	Complete named-state set required before CF-aware eligibility	Fail-closed tests and complete three-state ledgers	Missing-not-at-random state correction
Metamorphic invariance	Selected result equals T0 under irrelevant-relation, index/rebuild, and nullable-extension witnesses	180-question offline selector result	Value drift, paraphrase, production-state, or semantic robustness
Candidate-order stability	Original versus reversed stable-order tie breaking	Descriptive 101 versus 117–118/180 and 130/180 top ties	Not established; current selector is materially order-sensitive
Semantic validity	Agreement with frozen GridDB reference after board sealing	Finite historical-pool accuracy	Qualified human gold or field-operational correctness

Table 6. Versioned executor evidence matrix. A control is attributed only to the version and result ledger that actually used it.

Boundary	Controls/tests	Used for reported counts	Licensed evidence
Release v3	Read-only URI, `query_only`, authorizer, disabled extensions, opcode/time and row limits; 30 restored tests	5760 attempts and 80/100/101	Historical-pool descriptive selection and mutation/resource checks at v3 limits
FINAL additive	Function allowlist, projected-width, raw-cell-byte and total-result-byte budgets; 14 additive tests	None of 5760 or 80/100/101	Post-release control conformance only; non-retroactive
Formal-v5 scorer	Frozen 18-state evaluation path	25,920 rows and 66-item endpoint	Constructed-state agreement; not selector evidence
Superseded R3	Ten registered tests, no raw-BLOB accounting	No retained scientific count	Review provenance only

3.5. Inherited GridDB Factorial Design

The paired design crosses context package $c \in \{0,1\}$ and composite hint $h \in \{0,1\}$. F00 uses full DDL and global values without the hint; F01 adds the hint; F10 uses compact

Algorithm 1 Gold-isolated deterministic coordination

Require: question x_i, schema S_i, external candidates $\mathcal{Y}_i$, read-only executor, named state evidence

```
post structured intent and schema grounding to the append-only blackboard
canonicalize and deduplicate externally produced candidates
for each candidate y ∈ Y_i do
    apply single-statement read-only validation
    record execution and available reference-free state evidence
end for
if required named-state coverage is incomplete for a candidate then
    mark that candidate ineligible (fail closed)
end if
if no safe executable candidate meets all registered gates then
    abstain
else
    apply the fixed validation, state-evidence, coverage, and order rule
end if
post the decision and seal the blackboard
load reference results only for offline evaluation
```

domain-grounded context without the hint; and F11 combines compact context and the hint. The same 180 questions appear in all four cells for each backbone, yielding 720 Qwen and 720 Granite predictions.

Because each question appears in every prompt cell, cell differences are paired within question. The two backbone snapshots form parallel finite-corpus experiments rather than independent replications drawn from a model population. Difficulty labels and normalized-SQL clusters are retained as descriptive structure. Cluster-based randomization reduces the influence of near-duplicate SQL forms, while composition-sensitivity intervals show how estimates vary when observed clusters are resampled. Neither procedure converts this development-visible synthetic set into a population sample.

Qwen2.5-Coder-7B Q4_K_M and Granite-3.3-8B Q4_K_M were run at temperature zero with fixed snapshots. Each prompt produced exactly one response. The parser retained the first SQL candidate through the first semicolon; the validator then applied the read-only policy. There was no candidate ranking, multi-agent negotiation, or repair in this experiment.

Strict execution equality is primary. A common-target projected-column indicator is a secondary manipulation check. Paired finite-set contrasts use normalized-SQL clusters as dependence proxies, registered sign-flip tests, composition-sensitivity bootstrap intervals, and Holm correction. The intervals describe sensitivity to the observed corpus composition, not population confidence.

3.6. Component, Multi-State, and Public Protocols

The separately frozen component study contains 700 scored calls. E1 compares candidate generation with and without presented value evidence on 170 paired questions in 61 frozen groups. E2 requests up to three candidates and compares a deterministic reference-answer-independent selector with the first candidate on 180 questions in 70 groups. Question-level paired differences are averaged with equal question weights; bootstrap resampling and sign assignments operate at group level. Selection is sealed before gold loading. Rescue, harm, and oracle-at-three are descriptive; oracle-at-three is a gold-only upper-bound diagnostic. The three two-member Holm families are E1 across backbones, E2 across backbones, and the two cross-backbone modifiers.

The v5 reliability protocol reproduced all 1440 T0 labels and then executed every prediction over 18 states. The primary 66-question analysis uses a question-weighted logical-AND endpoint over 15 semantic states. A prediction passes only when it agrees with the reference on every included state. The nine registered contrasts form a separate Holm family; with only 12 structural groups, all $2^{12} = 4096$ group-sign assignments are enumerated. The sharp-null interpretation assumes exchangeability of those 12 group signs; the small group count limits randomization resolution, and the groups are dependence proxies rather than sampled deployment clusters.

The BIRD protocol compares direct prompting, decomposition, schema selection, and bounded execution repair. Each method has a fixed call pattern and zero retries. Accuracy and method differences retain equal item weighting over 500 items; pairwise randomization assigns one common sign to all paired item differences within each of 11 databases and applies Holm correction across 12 comparisons. Its sharp-null interpretation assumes database-level sign exchangeability. Eleven clusters give coarse attainable evidence and do not represent a random sample of power-grid databases.

The four protocols answer different questions and are not pooled. GridDB factorial cells test prompt-package manipulations on a controlled grid schema. E1 and E2 test presented values and reference-free selection under a prospectively frozen call-and-selection procedure on development-visible items. Formal-v5 tests agreement across constructed states. BIRD tests non-grid workflow portability. A positive result in one protocol cannot substitute for a missing end-to-end multi-agent comparison in another.

For every inferential family, multiplicity is defined before interpretation. The primary GridDB factorial family, the corrected projected-column family, the three component families, the nine multi-state contrasts, and the BIRD pairings remain separate because their endpoints and experimental units differ. Exact enumeration is used when the registered cluster count permits it; otherwise the retained composition-sensitivity procedure is reported. Effect sizes and complete counts remain visible even when a multiplicity-adjusted rule is not met.

3.7. Retrospective Offline Coordination Diagnostic

The replay verifies SHA-256 for two 720-row prediction ledgers and the 25,920-row formal-v5 ledger before reading them. For each question it pools the eight existing outputs from two backbones and four prompt cells, canonicalizes SQL, and removes duplicates. Frozen T0 fields provide execution and metadata-shape evidence. The Critic records counterfactual evidence as unavailable because existing state-agreement labels are gold-relative.

This is a coverage diagnostic. Candidates were generated under different models and prompt packages; the pool is not a deployable, matched multi-agent run. No selected output is compared with gold to produce an accuracy number.

3.8. Deterministic Descriptive Offline Re-Execution Study v3

The release-v3 study reuses exactly eight frozen source slots for each of the 180 GridDB evaluation questions: four Qwen prompt cells followed by four Granite prompt cells. It performs no LLM call and creates no new question or SQL label. Before any method selects, the three methods share one precomputed eight-by-four execution ledger: T0 plus three metamorphic witnesses for every slot. Thus 5760 is the shared physical evidence-collection count, not a separate natural runtime cost assigned to each selector. The fixed-order control then ignores the collected ranking evidence and chooses slot 0; validation-only passes an empty counterfactual mapping and ranks validation evidence only; the full condition requires complete 3/3 reference-free metamorphic invariance. Incomplete coverage or no passing candidate causes abstention.

The three query-blind witnesses have distinct SHA-256 values and are built from T0 without reading questions, prompts, predictions, scores, or gold. M1 adds an irrelevant relation and rows; M2 adds harmless indexes and rebuilds storage; M3 adds nullable probe columns to three business tables. M1 and M2 preserve original-schema denotation. M3 preserves explicitly projected original columns but deliberately allows wildcard projection to fail. Query order is preserved when candidate SQL contains `ORDER BY`; otherwise rows are compared as a multiset. The selector sees only question text extracted from frozen prompt ledgers, the schema, eight SQL strings, executor traces, and T0-to-witness equivalence. It does not see gold SQL, answer-shape metadata, correctness, difficulty, feature tags, required literals, or gold-relative formal-v5 labels.

These witnesses test metamorphic relations rather than alternative ground-truth answers. M1 asks whether an irrelevant table changes behavior, M2 whether logically neutral storage changes alter the result, and M3 whether projection is stable when nullable columns are appended. A pass means that the observed result satisfies the registered relation; it does not mean the original query expresses the user's intent. A fail can reveal brittleness, as with wildcard projection, but may also reflect a transformation outside the intended query equivalence class. For this reason, witness outcomes enter eligibility only under the disclosed construction and remain reference-free diagnostics.

Candidate order is a first-class experimental variable because the adjudicator uses stable order for equal scores. Original order places four Qwen slots before four Granite slots and follows the prompt-cell sequence within each backbone. Reverse-order sensitivity changes only this tie breaker after the same evidence has been collected. It is therefore a direct diagnostic of unresolved ranking information, not an alternative model or a new sample. The large change later observed under reversal motivates an abstention or clarification policy for future work.

Before v3 execution, a content-bound UTC timestamp, configuration, question-only prompt-derived selection view, two 720-row prediction ledgers, schema, T0, three witness databases, complete witness manifest, exact witness-builder bytes and SHA-256, executor, coordinator, core and release runners, all four regression-test files, and the passing pre-run test record were bound into a 21-*artifact* manifest with content SHA-256 `8b34bc370451173197ad07b460908537836409d6eb49f303caf46d13d53889e6`. The number 21 denotes manifest entries, not test cases. The restored frozen tree passes 30/30 tests across four files: 10 agent, 9 release-v3, 4 replay, and 7 executor tests. The superseded R3 additive suite passed its 10 registered tests but did not cover raw BLOB accounting; it is retained as review provenance rather than counted as a validated FINAL boundary. A transient 33/33 count produced while the frozen executor path had been overwritten was invalid, was reverted, and is excluded from the evidence record. The witness manifest independently binds its builder and three database hashes. Within each v3 run, all 180 blackboards and their decisions were sealed and written before that execution path directly loaded raw gold. Each run retains all 1440 source slots and all $180 \times 8 \times 4 = 5760$ shared attempts, including failures, without retry. Runs v3a and v3b produced byte-identical selection, evaluation, sensitivity, and summary files. Independent release audit passed mechanical integrity and descriptive traceability but failed the prior-outcome-independence evidence-class gate: the frozen pre-run test suite had opened v2 gold-derived evaluation rows for these same 180 items. The machine-readable `SUPERSESSION_NOTICE.json` controls interpretation and invalidates conflicting prospective fields in the immutable historical configuration, summaries, and internal 29/29 mechanical audit. V3 is therefore deterministic descriptive re-execution, not a scientific release independent of prior same-item outcomes. V1 and v2 remain preserved diagnostic predecessors and are excluded from claims.

Descriptive endpoints are strict execution accuracy over all 180 questions, coverage and abstention, complete-three-witness invariance of the selected candidate, and rescue/harm relative to fixed order. The retained sensitivity matrix crosses three weight policies, invariant-pass thresholds of 1/3, 2/3, and 3/3 while always requiring complete observation, and original versus reversed candidate-order tie breaking. Because same-item outcomes were accessible before v3, this matrix is reported only as descriptive analysis.

The later FINAL executor is a post-review engineering correction and is not part of the historical v3 test count. Its 14-test suite closes the raw-BLOB accounting defect found during R3 review, while the immutable R3 implementation and reviews remain preserved outside this FINAL candidate. No v3 result was rerun or reinterpreted with the corrected boundary.

3.9. Generative-AI Assistance and Author Responsibility

The local SQL-generation experiments used the explicitly identified Qwen2.5-Coder-7B-Instruct Q4_K_M and Granite-3.3-8B-Instruct Q4_K_M snapshots under the frozen protocols described above. Separately, during reconstruction and revision of this manuscript, the authors used the OpenAI Codex agent interface to propose and edit prose, suggest experimental and audit checks, review and draft Python/LaTeX code, assemble provenance and response matrices, and assist local build and visual-quality checks. The exact backend model identifier and version for every Codex session were not retained in the project records; this provenance boundary is reported explicitly, and no unrecorded model identifier is inferred.

All executable suggestions were run locally against retained inputs, and reported numerical results were taken from frozen or independently recomputed ledgers rather than accepted from generated prose. The authors retained responsibility for protocol authorization, evidence-class decisions, interpretation, citations, declarations, and the final manuscript. Codex output was not treated as qualified power-system expert annotation or adjudication. External machine-generated question–SQL candidates remain labeled machine silver and do not become human gold through model review. All six scientific figures are deterministic code-native graphics with file-level lineage; no generative-image model supplied quantitative evidence.

4. Results

No result in this section estimates the benefit of the five-role architecture relative to a call-matched single-agent or monolithic controller. RQ1 is answered only by software conformance: typed traces, fail-closed coverage, read-only enforcement, deterministic replay, and retained failures behaved as specified in the reported tests. The remaining subsections report inherited prompt/component experiments and outcome-exposed historical-pool diagnostics under their separate estimands.

4.1. Complete GridDB Cell Results

All 1440 scheduled predictions are present, and independent SQLite re-execution reproduced the stored execution verdicts. Table 7 reports strict execution equality.

Table 7. Strict execution equality over 180 GridDB questions per cell.

Backbone	F00 full/no hint	F01 full/hint	F10 compact/no hint	F11 compact/hint
Qwen	76/180 (0.4222)	129/180 (0.7167)	78/180 (0.4333)	108/180 (0.6000)
Granite	77/180 (0.4278)	100/180 (0.5556)	74/180 (0.4111)	108/180 (0.6000)

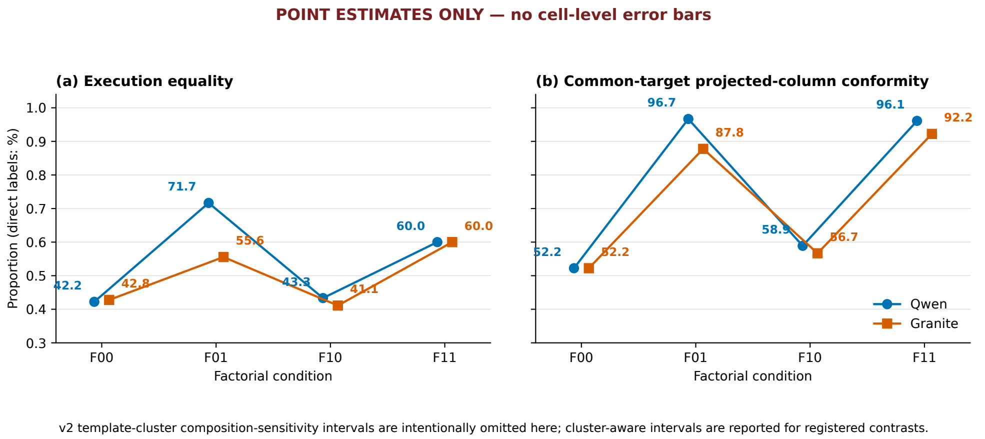

Figure 2. Audited GridDB cell point estimates. Each cell contains all 180 attempts; the figure is descriptive and does not replace registered factorial contrasts.

The common-target projected-column diagnostic is 0.5222 in F00 for both backbones. Qwen obtains 0.9667, 0.5889, and 0.9611 in F01, F10, and F11; Granite obtains 0.8778, 0.5667, and 0.9222. Returning the requested number of columns is therefore easier than returning the intended rows. In Qwen F01, projected-column conformity is 0.9667 while execution equality is 0.7167.

The eight cells also show why a single best prompt cannot represent the design. Adding the hint to full context increases the point estimate for both backbones, whereas adding compact context without the hint leaves execution equality close to F00. Under compact context, the hint produces 108 correct answers for both models, but this numerical convergence arises from different unhinted baselines and does not establish identical behavior. Complete cell reporting preserves these interactions rather than selecting only the highest cell.

The 0.2500 gap between Qwen F01 projected-column conformity and strict execution equality warns against structural proxy endpoints. A model may recognize that one column, an aggregate, or an ordered row is requested while still choosing the wrong table, join, predicate, value, or boundary. Shape diagnostics can localize errors and inform adjudication, but they cannot replace result equality or qualified semantic review.

4.2. Factorial Effects and Multiplicity

For Qwen, the composite-hint execution effect is +0.2306, the package–hint interaction is -0.1278, and the compact-package main effect is -0.0528. Granite's corresponding estimates are +0.1583, +0.0611, and +0.0139. Zero of nine primary execution tests survives Holm correction. Qwen's hint effect has raw/adjusted $p = 0.01372/0.09604$, its interaction 0.00919/0.07352, and the cross-backbone hint modifier 0.00640/0.05760. No primary factorial execution effect or modifier is therefore described as statistically nonzero.

The normalized-SQL mapping contains 70 groups, 58 singletons, and a maximum group size of 19. Question-level weighting is retained; groups are resampling and sign-assignment proxies, not verified authoring templates or sampled clusters. The randomization interpretation requires exchangeability of group signs under the tested sharp null, while the composition interval asks how the finite-corpus mean changes when those observed groups are reweighted. Neither assumption follows from the data, and the high singleton fraction limits protection against unmeasured dependence.

The displayed composition intervals are pointwise sensitivity intervals; they are neither simultaneous/familywise intervals nor inversions of the nine Holm-adjusted randomization tests. An interval that excludes zero can therefore coexist with a Holm non-rejection.

Table 8. Complete primary factorial family for strict execution equality. Estimates are question-weighted finite-set differences; intervals are 95% normalized-SQL-group composition-sensitivity intervals. Each sign-randomization used 70 groups, 100,000 draws, and the displayed raw values were Holm-adjusted jointly over all nine rows.

Scope	Effect	Estimate	Composition interval	Raw p	Holm p
Qwen	Compact-package main	-0.0528	[-0.1131, 0.0000]	0.10523	0.52614
Qwen	Structural-hint main	0.2306	[0.0548, 0.4355]	0.01372	0.09604
Qwen	Package × hint	-0.1278	[-0.2454, -0.0341]	0.00919	0.07352
Granite	Compact-package main	0.0139	[-0.0584, 0.0798]	0.85021	1.00000
Granite	Structural-hint main	0.1583	[-0.0193, 0.3662]	0.15480	0.61919
Granite	Package × hint	0.0611	[-0.0539, 0.2013]	0.55890	1.00000
Granite–Qwen	Backbone × package	0.0667	[-0.0360, 0.1814]	0.41478	1.00000
Granite–Qwen	Backbone × hint	-0.0722	[-0.1229, -0.0276]	0.00640	0.05760
Granite–Qwen	Three-way interaction	0.1889	[0.0090, 0.4321]	0.07010	0.42060

The multiplicity decision, not interval exclusion, controls statements about the registered nine-test family.

No minimum practically important execution-accuracy difference, equivalence margin, or non-inferiority margin was registered. Therefore the study cannot infer practical equivalence from a non-rejection, and the 1/180 complete-witness difference is only a mechanism trace rather than a meaningful-benefit result.

In the separately corrected secondary family, the common-target hint effects survive: +0.4083 for Qwen (adjusted $p = 0.000090$) and +0.3556 for Granite (adjusted $p = 0.01944$). Because the hint supplies structural information later scored by this endpoint, these are manipulation-check results, not independent semantic evidence.

The multiplicity result changes the emphasis of the factorial study. Large raw contrasts motivate hypotheses about structural instructions, but none of the nine primary execution comparisons meets the registered familywise rule. The projected-column findings verify that the manipulation changed a closely aligned surface property. They do not show that the compact package or hint produces a backbone-independent accuracy gain. Future work should separate schema pruning, value presentation, normalization, and structural hints rather than retaining composite packages.

4.3. Component Results

The component run completed all 700 scored calls under a prospectively frozen call-and-selection procedure on development-visible GridDB items. On the paired 170-question subset, Qwen first-candidate correctness changed from 83/170 to 101/170, a +0.1059 difference (95% composition-sensitivity interval [+0.0282, +0.2013], adjusted $p = 0.0310$). Granite remained 69/170 in both paired conditions, with an interval spanning zero. The cross-backbone modifier did not meet the registered rule, so the Qwen result is not replicated across backbones and is not a five-role result.

The deterministic selector changed 24 Qwen and 36 Granite choices. Relative to the first candidate, it rescued eight and harmed one Qwen question, and rescued ten while harming none for Granite. Execution effects were +0.0389 and +0.0556; neither survived its two-test Holm family. Oracle-at-three values of 0.6389 and 0.4667 are gold-only diagnostics and are not deployable selector results.

For Qwen, the value-evidence intervention adds 18 correct first candidates when the 170 eligible V0 questions are compared with their V1 counterparts under the registered analysis; the corresponding Granite estimate is zero. The disagreement is more informative than averaging the backbones: the effect depends on how a frozen model uses schema and value context. Oracle-at-three measures candidate-pool headroom, whereas the reference-free selector measures how much of that headroom the recorded rule can recover without gold.

Table 9. Complete component endpoints from the 700-call audited component protocol whose calls and selection were prospectively frozen on development-visible GridDB items. V0 and paired V1 contain the same 170 value-intervention-eligible questions; the all-item V1 rows contain 180. These are component, not five-role, outcomes.

Backbone	Condition	First correct	Selector correct	Oracle@3 diagnostic
Qwen	V0 no value evidence	83/170 (0.4882)	83/170 (0.4882)	100/170 (0.5882)
Qwen	V1 with evidence, paired subset	101/170 (0.5941)	—	—
Qwen	V1 with value evidence	105/180 (0.5833)	112/180 (0.6222)	115/180 (0.6389)
Granite	V0 no value evidence	69/170 (0.4059)	69/170 (0.4059)	72/170 (0.4235)
Granite	V1 with evidence, paired subset	69/170 (0.4059)	—	—
Granite	V1 with value evidence	71/180 (0.3944)	81/180 (0.4500)	84/180 (0.4667)

Table 10. Complete component-effect families. Intervals are pointwise group-composition sensitivity summaries, not simultaneous intervals and not inversions of the Holm tests. Each family contains the two displayed members indicated by its identifier.

Family	Contrast	N	Groups	Estimate	Composition interval	Raw p	Holm p
E1	Qwen V1–V0 first candidate	170	61	0.1059	[0.0282, 0.2013]	0.01552	0.03104
E1	Granite V1–V0 first candidate	170	61	0.0000	[-0.1902, 0.1705]	1.00000	1.00000
E2	Qwen selector–first V1	180	70	0.0389	[-0.0081, 0.1071]	0.50080	0.50080
E2	Granite selector–first V1	180	70	0.0556	[0.0075, 0.1232]	0.06425	0.12850
Cross-E	Granite–Qwen E1 effects	170	61	-0.1059	[-0.2701, 0.0394]	0.26986	0.53971
Cross-E	Granite–Qwen E2 effects	180	70	0.0167	[-0.0062, 0.0457]	0.37477	0.53971

The three retained two-member Holm families are therefore E1 across the two backbones, E2 across the two backbones, and Cross-E across the two backbone modifiers. The Granite E2 pointwise composition interval excludes zero while its family-adjusted decision does not meet the registered rule; these are different summaries and are not treated as contradictory. The 61- and 70-group mappings retain question-level weighting and assign randomization signs at the frozen group level. As in the factorial analysis, these observed groups are dependence proxies rather than a probability sample, so exchangeability of group signs is an assumption rather than an empirical result.

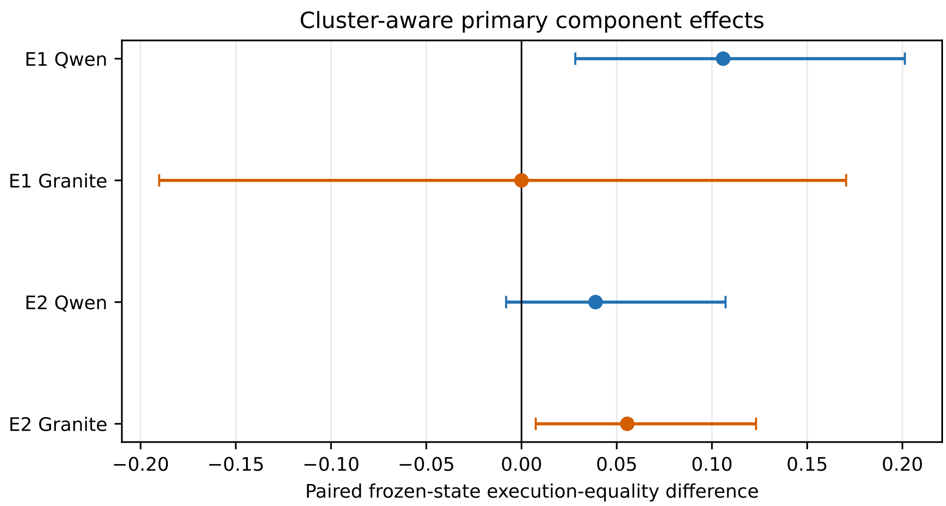

Figure 3. Component effects under a prospectively frozen call-and-selection procedure on development-visible GridDB items. Intervals describe composition sensitivity; the registered decision also requires multiplicity-adjusted evidence.

4.4. Multi-State Reliability Results

The v5 scorer produced all 25,920 registered rows, and its T0 slice reproduced 1440/1440 snapshot labels. Across the 66-question automatic subset, 15-state logical-AND rates range from 0.6212 (Granite F01) to 0.8182 (Granite F00). All nine Holm-adjusted values equal 1.0000 under exact cluster-sign enumeration. No context, hint, interaction, or cross-backbone multi-state effect meets the declared rule. These rates characterize agreement over constructed witness states, not population accuracy or human-certified equivalence.

The ordering under this endpoint differs from the single-snapshot table. Granite F00 has the highest logical-AND rate on the 66-question automatic subset even though Granite F11 has the larger T0 execution count over all 180 questions. The denominators and endpoints differ, so this is not a contradiction. It demonstrates why a robustness statement must name both the admitted question subset and the required state set.

Table 11. Complete 15-state logical-AND counts for the prediction-blinded 66-question automatic subset.

Backbone	F00	F01	F10	F11
Qwen	49/66 (0.7424)	48/66 (0.7273)	53/66 (0.8030)	43/66 (0.6515)
Granite	54/66 (0.8182)	41/66 (0.6212)	51/66 (0.7727)	49/66 (0.7424)

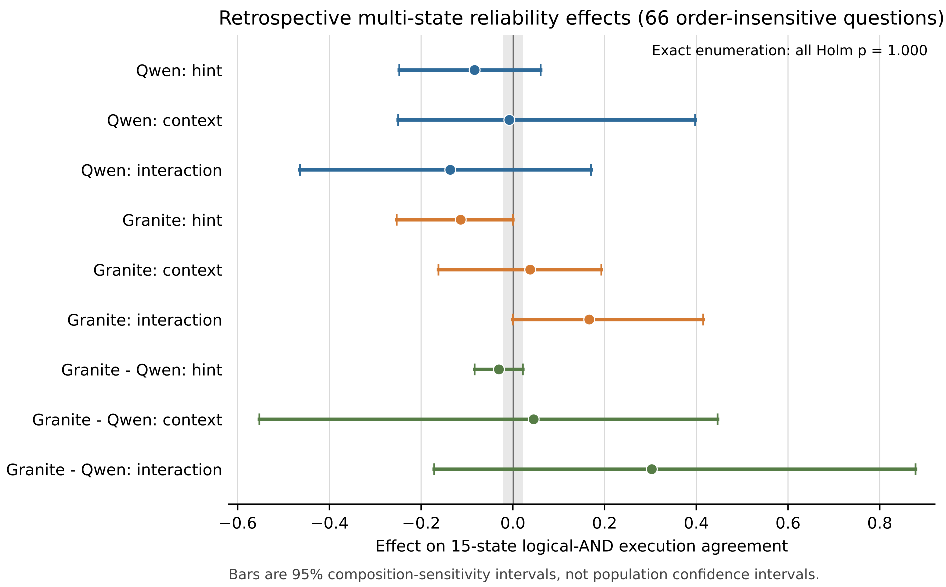

Figure 4. GridDB multi-state logical-AND agreement for the prediction-blinded automatic subset. The endpoint is a constructed-state robustness diagnostic.

4.5. Public BIRD Results

Table 12 reports all eight cells. BIRD Mini-Dev is a public non-grid portability benchmark and does not validate power-grid semantics. B0–B2 use one physical generation call per item; B3 uses two, so the inherited comparison estimates frozen workflow efficacy under unequal call counts rather than a call-matched repair effect. Each backbone completed 2500 calls, 2000 final predictions, zero retries, and zero final-ledger omissions; latency and token totals were not promoted into the retained summary and are therefore reported as unavailable rather than reconstructed.

After Holm adjustment, Qwen decomposition is 0.076 below direct prompting (adjusted $p = 0.0430$), and schema selection is 0.092 above decomposition (adjusted $p = 0.0117$). No Granite contrast survives correction. Different method orders across

Table 12. Inherited BIRD Mini-Dev v1.1 results (non-grid portability evidence; 11 database clusters). Intervals are database-cluster composition-sensitivity intervals.

Backbone	Method	Correct/500	Accuracy	Interval	Calls/item	Final-ledger omissions
Qwen	B0 direct	189/500	0.378	[0.286, 0.470]	1	0
Qwen	B1 decomposition	151/500	0.302	[0.203, 0.401]	1	0
Qwen	B2 schema selection	197/500	0.394	[0.279, 0.511]	1	0
Qwen	B3 execution repair	174/500	0.348	[0.250, 0.452]	2	0
Granite	B0 direct	102/500	0.204	[0.125, 0.284]	1	0
Granite	B1 decomposition	105/500	0.210	[0.131, 0.294]	1	0
Granite	B2 schema selection	101/500	0.202	[0.137, 0.273]	1	0
Granite	B3 execution repair	118/500	0.236	[0.159, 0.318]	2	0

the two model snapshots argue against a universal prompting recipe. The two failed incident runs totaling 2476 physical calls remain excluded and immutable; they are not retries contributing to these denominators.

The BIRD results also constrain the meaning of "agentic" improvement. Decomposition reduces Qwen accuracy relative to direct prompting, while bounded repair uses twice the physical calls without becoming the best Qwen method. Granite shows a different descriptive ordering and no adjusted contrast. More stages or calls therefore do not guarantee a better outcome. A future comparison must count physical generations, retries, context tokens, and failed attempts, not only final predictions.

4.6. Retrospective Replay Coverage

The replay audited all 180 GridDB questions without a generation call. At least two distinct frozen SQL candidates were present for 173 questions. After safety checks and binding to consistent T0 execution evidence, 172 questions had at least two eligible candidates and entered retrospective adjudication. Seven failed closed because only one unique SQL remained, and one failed because fewer than two candidates were eligible.

Table 13. Retrospective offline coordination diagnostic coverage. Counts are not accuracy and do not estimate multi-agent improvement.

Coverage gate	Questions
Audited	180
At least two unique frozen SQL candidates	173
At least two eligible candidates	172
Retrospectively adjudicated	172
Failed closed: one unique candidate	7
Failed closed: fewer than two eligible candidates	1
Reference-free counterfactual evidence available	0

Unique candidate counts ranged from one to eight: 1 (7 questions), 2 (19), 3 (15), 4 (18), 5 (27), 6 (23), 7 (45), and 8 (26). These counts show that frozen assets can populate the coordination interfaces for most questions. They do not show that retrospectively selected SQL is correct. No retrospective accuracy, execution gain, rescue rate, or multi-agent superiority is reported.

4.7. Descriptive Offline Coordination-Selection Results

The deterministic descriptive re-execution retained 5760 candidate–state attempts; 332 were non-executable or authorization/resource/SQL failures and remained in the all-attempt ledger. Every question nevertheless had at least one eligible candidate, so both adjudicating selectors covered 180/180 questions and abstained on 0/180. This zero-abstention result is a consequence of at least one eligible slot plus forced stable-order resolution of post-ranking ties, not confidence-calibrated coverage. Table 14 reports the recorded rules.

Keeping all 332 failures matters for two reasons. Removing them would overstate the executability of the historical pool, and retaining their slot identities verifies that selectors did not receive replacements or retries. Full question-level coverage means only that another slot was eligible for every question; it does not mean every candidate was valid or that an operational system should always answer.

Table 14. Descriptive release-v3 re-execution over one historical eight-slot pool per question. All methods choose after the same precomputed candidate/evidence ledger of 32 attempts per question. Accuracy is finite-corpus strict execution equality after all blackboards were sealed; this is not a new-generation or five-role end-to-end experiment.

Selector	Correct/180	Accuracy	Covered	Abstained	Invariant	Rescue/harm
Fixed order, shared ledger	80/180	0.4444	180	0	177/180	0/0
Validation rank, shared ledger, no CF	100/180	0.5556	180	0	179/180	22/2
Complete-three-witness selector	101/180	0.5611	180	0	180/180	23/2

Figure 5 juxtaposes the two recorded endpoints. Validation-only and complete-witness selection differ by one frozen-reference match and one invariant selection. This small incremental change contrasts with their 20–21-match difference from fixed order and focuses interpretation on validation ranking and tie policy rather than on a broad counterfactual benefit.

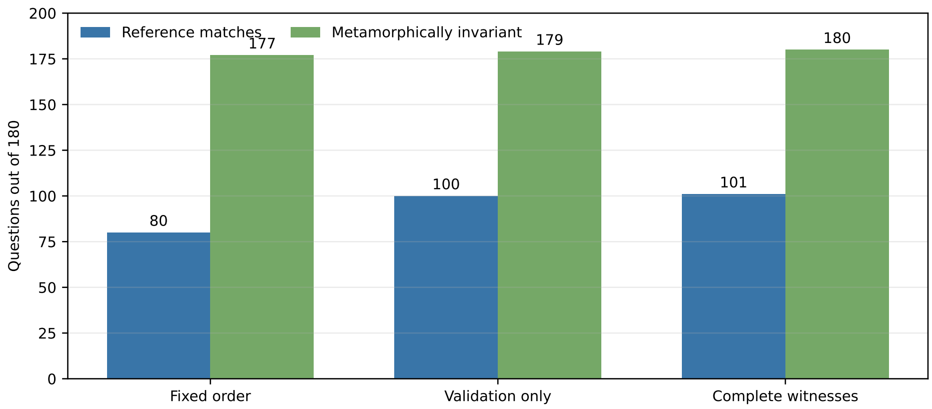

Figure 5. Descriptive release-v3 selector endpoints over 180 historical-pool questions. Reference matches are loaded only after board sealing; invariance denotes agreement across the three constructed witnesses. Bars are not a prospective multi-agent comparison.

The two adjudicating selectors each had more than one highest-scoring candidate on 130/180 questions (Table 15). Their mean top-tie multiplicities were 5.4000 and 5.3889. All eight candidates tied on 79 validation-only questions and 78 complete-witness questions. With original-order tie breaking, 178/180 selections came from Qwen source slots; this reflects source order, not a learned preference or semantic advantage.

Thus, most apparent source preference is mechanically induced. When all eight candidates tie, the first slot is selected regardless of backbone quality; when seven tie, the earliest surviving slot has the same advantage. The observed 178 Qwen selections cannot be interpreted as an adjudicator comparison between Qwen and Granite. It instead measures how often the available reference-free features failed to separate candidates before the stable-order rule acted.

Table 15. Complete top-score tie diagnostic for the two adjudicating selectors. Multiplicity distributions are reported as "tie size:number of questions".

Selector	Tied questions	Mean size	Multiplicity distribution	Selected source-slot counts
Validation only	130/180	5.4000	1:50; 2:5; 3:3; 4:4; 5:5; 6:8; 7:26; 8:79	Qwen F00/F01/F10/F11: 115/49/12/2; Granite F00/F10: 1/1
Complete witnesses	130/180	5.3889	1:50; 2:5; 3:3; 4:4; 5:5; 6:9; 7:26; 8:78	Qwen F00/F01/F10/F11: 114/50/12/2; Granite F00/F10: 1/1

Validation-only ranking added 20 net correct items relative to fixed order through 22 rescues and 2 harms. Complete-witness selection changed the chosen candidate on only Q039, yielding one further frozen-reference match (101/180 versus 100/180), one additional rescue, no additional harm, and no coverage or abstention change. Table 16 gives the complete constructed trace.

Table 16. Q039 is the sole validation-only versus complete-witness selection difference. Correctness is agreement with the frozen synthetic reference, not an independently adjudicated engineering judgment.

Field	Validation-only	Complete witnesses	Interpretation
Natural-language request	"Show the first scheduled work order after June 2024."		Ambiguous: "after June" can imply after 30 June, and "scheduled" can denote a status or merely a scheduled date.
Selected query	`SELECT * FROM work_orders WHERE scheduled_date > '2024-06-01' ORDER BY scheduled_date ASC LIMIT 1`	`SELECT work_order_id, scheduled_date FROM work_orders WHERE scheduled_date > '2024-06-01' ORDER BY scheduled_date ASC LIMIT 1`	Both omit an explicit scheduled-status predicate and use a date boundary whose intended meaning lacks expert adjudication.
M3 outcome	Fails when nullable columns extend wildcard width	Preserves the explicit two-column projection	Constructed projection-stability behavior
Frozen-reference match	Incorrect	Correct	Outcome-exposed 1/180 trace only

M3 added nullable columns to `work_orders`; consequently, complete-witness eligibility rejected the wildcard projection and retained the explicit `work_order_id, scheduled_date` projection. Because neither query resolves the request's status and date-boundary ambiguity through qualified review, this observation is not called an engineering correctness rescue. It is a synthetic, outcome-exposed projection-stability trace, not evidence of general counterfactual reasoning, semantic validity, robustness, coordination, or five-role benefit.

Table 17 reports all 18 sensitivity cells. Every original-order cell produced 101/180; reverse order produced 117/180 under default and intent-emphasis weights and 118/180 under validation-equal weights. These more favorable values are outcome-exposed diagnostics, not preferred alternatives. The 16–17-item shift and high tie multiplicity show that the apparent selector behavior is materially order-dependent.

Threshold invariance across 1/3, 2/3, and 3/3 occurs because the selected candidates' recorded witness patterns and score ties do not change the final original-order choices under these policies. It does not show that threshold selection is generally unimportant. Conversely, the reverse-order increase cannot justify choosing reverse order after seeing gold. Together, the cells diagnose inadequate discriminative evidence and define a pre-registration requirement for the next pool.

Table 17. All 18 complete-witness selector sensitivity cells. Each row contains the original- and reverse-order cells for one weight/threshold combination; coverage is 180/180 throughout.

Weight policy	Minimum witness passes	Original order correct/180	Reverse order correct/180
Default	1/3	101 (0.5611)	117 (0.6500)
Default	2/3	101 (0.5611)	117 (0.6500)
Default	3/3	101 (0.5611)	117 (0.6500)
Intent emphasis	1/3	101 (0.5611)	117 (0.6500)
Intent emphasis	2/3	101 (0.5611)	117 (0.6500)
Intent emphasis	3/3	101 (0.5611)	117 (0.6500)
Validation equal	1/3	101 (0.5611)	118 (0.6556)
Validation equal	2/3	101 (0.5611)	118 (0.6556)
Validation equal	3/3	101 (0.5611)	118 (0.6556)

The full machine-generated numerical supplement binds all eight GridDB cells, 12 component endpoints, eight 15-state cells, 18 sensitivity cells, the 360-row item-level tie ledger, selected-origin counts, Q039 trace, source paths, hashes, and byte counts in `R3_EVIDENCE_TABLES.json` and companion CSV files. Table 12's final column records only omissions from the accepted final ledgers; it is not a taxonomy of parsing, authorization, execution, provider, or incident failures. The two excluded BIRD incidents are reported separately above. This result supports auditability and exposes a design risk; it does not establish a generally robust or superior selector.

Across the results, the strongest positive numbers occur under different evidence classes: Qwen F01 is the best GridDB generation cell, schema selection is the best Qwen BIRD workflow, complete-witness original-order selection yields 101 historical-pool matches, and reverse order raises the descriptive count to 117–118. These values cannot be combined into one framework accuracy because they use different questions, candidates, calls, visibility histories, and selectors. Their joint value is diagnostic: they identify components for a clean prospective protocol and claims that remain unsupported.

5. Discussion

5.1. What the Evidence Supports

The combined evidence supports four bounded findings. First, explicit structural and SQL-operation instructions strongly alter projected-column adherence, but surface conformity does not establish correct execution. Second, the 700-call component experiment, whose call-and-selection procedure was prospectively frozen on development-visible items, finds that presented value evidence matters for one frozen Qwen snapshot while showing no effect for Granite; this is component evidence, not a five-role outcome. Third, candidate selection and prompting are backbone-dependent: the component selector did not meet its registered efficacy rule, and the descriptively best BIRD method differed between Qwen and Granite. Fourth, on the historical eight-slot pool, validation ranking changed the observed finite-set score relative to fixed order, but complete witness evidence changed only Q039, and top-score ties on 130/180 items made results strongly dependent on arbitrary source order.

These findings justify separating intent, schema grounding, synthesis, validation, and state evidence. They do not establish that five roles are superior to one call or that newly generated multi-agent candidates would reproduce the offline gain. The architecture's demonstrated value is methodological and safety-bounded: typed contracts, database-enforced eligibility, complete-evidence gates, deterministic selection, abstention, incident retention, and a sealed gold boundary make later comparisons reproducible and falsifiable.

Figure 6 maps each evidence source to its claim ceiling. The map prevents results from being promoted merely because they appear in the same implementation: executor tests establish a software boundary, GridDB and component studies establish finite-corpus behavior, BIRD establishes non-grid portability, and v3 establishes descriptive selection on a previously evaluated pool. Prospective five-role benefit and operational validity remain untested.

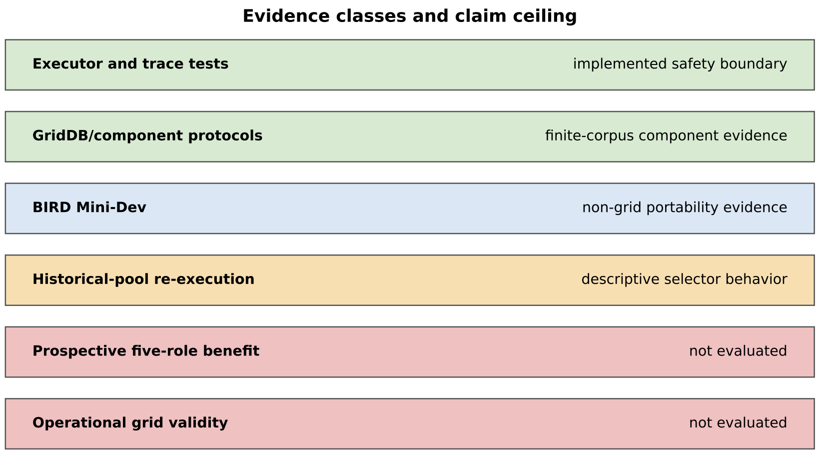

Figure 6. Evidence classes and permitted claim ceiling. Colors distinguish audited or bounded evidence, non-grid portability, descriptive prior-outcome-exposed analysis, and evaluations not performed.

5.2. Interpreting the Retrospective Replay

The replay asks whether inherited ledgers contain enough distinct executable candidates to exercise the coordination interfaces. For 172 of 180 questions, they do. This reduces implementation risk and identifies eight fail-closed cases that require a registered fallback or a new matched generation policy.

The replay cannot estimate a coordination effect. Its pool mixes two backbones and four prompt packages, so generation condition and diversity are confounded. It also lacks reference-free counterfactual equivalence evidence. Post hoc gold scoring of selected outputs would remain exploratory and cannot substitute for a prospective matched comparison.

5.3. Interpreting the Descriptive Re-Execution

The release-v3 study hash-binds the selection view, rules, code, tests, and state inputs and verifies the software fact that each execution path directly loaded raw gold only after all blackboards were sealed. This does not establish outcome blindness: its frozen test suite had accessed gold-derived v2 evaluation rows for the same items before v3. The additive supersession notice is the first authority for interpreting the immutable release package and invalidates its historical prospective, prespecified, and outcome-unseen labels. The study is therefore a deterministic descriptive re-execution over a historical candidate pool. Each selector inherited the same precomputed eight-slot candidate/evidence ledger; this does not mean equal historical model calls between source conditions, and it does not turn those candidates into contemporaneous agent outputs.

The small contrast between validation-only and complete-witness selection is a mechanism trace, not a gain estimate. V3 uses three different query-blind database hashes and exercises an irrelevant-relation witness, an index/storage witness, and a nullable schema-extension witness. These are constructed operator families rather than independent samples from a deployment population. The full gate changes only Q039 because

M3 exposes wildcard-projection arity while preserving explicit-column projections. The natural-language request itself admits unresolved status and date-boundary interpretations, so the frozen-reference match cannot be promoted to power-grid semantic correctness. An earlier v1 diagnostic used three same-hash insertion files and is retained as an excluded incident; v2 is retained as a provenance-incomplete predecessor. The supersession notice excludes their prospective interpretations and none of these predecessors supports the current scientific claim.

The tie audit quantifies a second limitation: 130/180 questions have multiple highest-scoring candidates for each adjudicating selector, with approximately 5.4 candidates in the average top-tie set. The original-order rule consequently chooses source slots without semantic evidence and selects Qwen-origin candidates on 178/180 items. Reporting only the 0.5556 default or choosing the 0.6556 reverse-order sensitivity after outcome access would be misleading. A later rule-development study requires development-only calibration, richer reference-free evidence, and a genuinely untouched evaluation resource.

5.4. Power-Grid Use and Reproducibility

Database-enforced read-only execution, resource limits, syntactic validity, and result-shape conformity are necessary safeguards, not operational certification. A safely executed query can still return the wrong assets, omit time bounds, misinterpret codes, or mishandle ties. Table/column/function allowlists constrain SQL operations inside this local evaluator; user authorization requires a separately authenticated identity, institutional access policy, and data-owner decision. The additive FINAL raw-cell-byte, budgeted-result-byte, and output-width controls were not used in release v3 and cannot be invoked to reinterpret its results. They bound returned values within one Python/SQLite process, not SQLite's entire temporary-memory or process-memory footprint. Multi-state witnesses reduce accidental snapshot equality but cannot cover all distributions or schema changes. Generated SQL and returned source rows require authorized human inspection before any maintenance, protection, dispatch, or asset-management decision.

Negative results are part of the contribution. Zero of nine primary factorial execution tests survives correction. The deterministic component selector meets its efficacy rule for neither snapshot. No registered multi-state effect survives correction. Granite and Qwen do not share the same best BIRD method. Immutable ledgers, direct re-execution, registered test families, portable manifests, and explicit incidents preserve these outcomes rather than collapsing them into a success narrative.

6. Limitations and Future Work

The primary domain corpus is one synthetic database with eight tables, 98 rows, and 180 evaluation questions. Its evaluation partition was visible during development. The 70 structural groups are dependence proxies, and 58 are singletons. Results cannot be generalized to production schemas, permissions, missing-data patterns, units, access-control policies, or workloads.

Only two quantized local snapshots were evaluated. Model family, size, quantization, serving software, and serialization may interact. The compact package bundles schema reduction, values, normalization, and serialization; the hint bundles projection, aggregation, grouping, ordering, and limit rules. The inherited factorial experiment identifies package effects rather than contributions of individual internal rules.

The five-role core and frozen release-v3 executor pass 30 unit/adversarial/regression tests; the separate FINAL executor passes 14 additive tests, including four new BLOB-accounting and trace/boundary methods. The 21-entry freeze manifest is not a test count, and the invalid transient 33/33 run is excluded. The system has not completed a new-generation experiment on an evaluation resource whose outcomes were previously inaccessible to the study process. Its Analyst and Cartographer are deterministic skeletons, its Synthesizer packages external candidates, and its adjudication weights have not been independently optimized. Release v3 descriptively re-executes selection over historical candidates and previously evaluated items; it does not estimate generation, communication, latency, token, or autonomous-agent effects. Existing formal-v5 agreement labels are gold-relative and intentionally excluded from selection.

RTS-GMLC, SimBench, and NERC question–SQL assets remain machine-adjudicated silver data. Qualified review is required for intended projections, units, ordering, tie handling, and result granularity before external accuracy is reported.

Future work should execute a new hash-locked, call-matched comparison of four contemporaneous generation conditions: a single direct candidate; staged question/schema handoffs with one candidate; a fixed multi-candidate budget with validation and adjudication but no counterfactual evidence; and the same candidate pool with a pre-registered multi-family reference-free state suite. Model snapshot, question order, database state, decoding, candidate count, physical calls, and evaluator must be held constant. The protocol must pre-register execution accuracy, valid SQL, abstention, tokens, latency, failure taxonomy, clustered inference, multiplicity, and incident retention. The candidate-order rule must be chosen on development-only evidence before the untouched evaluation set is opened.

Further work can replace lexical grounding with a trained schema linker, add a low-confidence full-schema fallback, compare deterministic adjudication with budget-matched voting, and evaluate risk–coverage calibration. Qualified power-system/database reviewers must independently assess domain semantics; LLM or author-assisted triage cannot be relabeled as independent expert gold. Every change should receive a new protocol identity rather than modifying a completed run.

7. Conclusions

MA-SQLGrid reframes power-grid text-to-SQL as an auditable coordination problem. Five typed software roles separate question analysis, schema mapping, candidate packaging, safety and execution validation, and named-state criticism. A deterministic controller records their handoffs, selects only eligible candidates, and abstains when required evidence is incomplete. Here, *multi-agent* denotes this inspectable role decomposition, not demonstrated benefit from five autonomous language-model agents. The frozen executor enforces read-only and resource limits at the SQLite boundary rather than relying on lexical filtering alone; the separate FINAL variant adds tested raw-cell-byte, budgeted-result-byte, width, and function controls that are not retroactive to v3 and are not whole-process memory isolation.

The inherited experiments provide a substantial but bounded evidence base. The 1440-prediction GridDB study shows strong projected-column instruction uptake but no primary execution effect surviving Holm correction. The 700-call component study supports presented values for one Qwen snapshot, not Granite, and does not validate its selector under the registered rule. The 25,920-row state study finds no corrected factorial agreement effect. The 5000-call BIRD protocol supplies a non-grid public comparison whose best method depends on the backbone and whose methods use unequal call counts. The retrospective replay confirms candidate coverage for 172 of 180 questions without accuracy claims.

The descriptive offline re-execution records 80/180 for fixed order, 100/180 for validation-only, and 101/180 for complete witnesses. Both adjudicating selectors have highest-score ties on 130/180 questions, average tie multiplicity is approximately 5.4, and reverse order changes the complete-witness sensitivity result to 117–118/180. The sole Q039 difference is a constructed, outcome-exposed, semantically ambiguous projection trace. These facts, the historical mixed-source pool, development-visible GridDB, and absent qualified semantic review prevent a general multi-agent superiority or power-grid deployment claim. The original title is retained as the framework identity; *robust* refers only to explicitly tested mutation-denial, bounded-execution, complete-evidence, and three-witness mechanisms, not universal or operational robustness.

Supplementary Materials: The FINAL editor/reviewer package is generated from an explicit allowlist and clean-extraction checked. It contains the exact manuscript source and PDF, bibliography, all six figure artifacts and lineage sources, complete numerical tables, 360-row tie ledger, Q039 trace, supersession notice, corrected FINAL executor and tests, three R3 reviews, the itemized FINAL response matrix, rights inventory, audit reports, and hashes. It excludes old-title manuscripts, old PDFs, `ROUND1_SECTION_DRAFT`, credentials, caches, and excluded incident outputs presented as evidence. This local package does not close repository synchronization, third-party permission, qualified expert review, or untouched end-to-end experimental gates. The public repository at https://github.com/gaoxingkele/ma-sqlgrid must still be synchronized and tagged before submission.

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Data Availability Statement: The public repository at https://github.com/gaoxingkele/ma-sqlgrid must be synchronized and tagged before submission; this statement does not assert that its current contents match the FINAL candidate. BIRD Mini-Dev remains available from its public provider. Owing to third-party licensing restrictions, restricted source records and source-dependent RTS-GMLC or SimBench artifacts may be requested from the corresponding author for editorial and peer-review verification, subject to third-party permission and applicable licenses; reviewer access is not a redistribution license. The local allowlist package intentionally excludes source assets whose redistribution or reviewer-sharing permission is unresolved. Its rights inventory distinguishes public, restricted, derived, machine-silver, historical, and excluded-incident materials and binds the supersession notice, numerical supplement, code, tests, and independent audits.

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